Exercises week 3 – Answers

8CM10 - Introduction to Quantum Physics and Structure of Matter

Schrödinger Equation

1. Consider a (one-dimensional) particle that moves in a constant potential energy $U(x) = c$ for all x.
    a. Solve the Schrödinger equation for this case. Show that there are only bounded solutions $\Psi(x)$ if the total energy E satisfies $E \geq c$.
    b. What is the wavelength of $\Psi(x)$ for a given value of E, with $E \geq c$?
    c. What is the de Broglie wavelength of a particle with a kinetic energy $K = E - c$ for a given value of E, with $E \geq c$? Compare this value to the value you found for (b).
    d. What is the physical difference between this particle and a free particle?

(a) The Schrödinger equation for a particle with mass m in a potential $U(x)$ can be written as:

$$\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2}(E - U(x))\Psi = 0 \ .$$

If the potential $U(x)$ is equal to a constant c this becomes

$$\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2}(E - c))\Psi = 0 \ .$$

If we are looking for solutions with energy $E \geq c$ this can be rewritten as

$$\frac{d^2\Psi}{dx^2} = -k^2\Psi \quad \text{with} \quad k^2 = \frac{2m}{\hbar^2}(E - c) \geq 0.$$

This equation has general solutions

$$\Psi(x) = A\sin(kx) + B\cos(kx) \ .$$

Note that if $(E - c) < 0$ so $E < c$, then the equation becomes

$$\frac{d^2\Psi}{dx^2} = a^2\Psi \quad \text{with} \quad a^2 = -\frac{2m}{\hbar^2}(E - c) \geq 0.$$

This equation has general solutions

$$\Psi(x) = Ae^{ax} + Be^{-ax}.$$

These equations explode for large positive or negative x so these are not bounded solutions.

(b) The wavelength λ of these solutions of the Schrödinger equation is given by

$$\lambda = 2\pi/k = \frac{2\pi}{\sqrt{\frac{2m}{\hbar^2}(E - c)}} = \frac{2\pi\hbar}{\sqrt{2m(E - c)}} = \frac{h}{\sqrt{2m(E - c)}} \ .$$

(c) The De Broglie wavelength of a particle with momentum p is given by $\lambda = h/p$. The momentum is related to the kinetic energy K by $p = \sqrt{2mK}$, where m is the mass of the particle. Hence

$$\lambda = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2m(E - c)}} \ .$$

So for a constant potential the solution $\Psi(x)$ of the Schrödinger equation has precisely the wavelength given by De Broglie.

(d) Since the potential $U(x)$ is constant, there is no force acting on the particle. Hence there is no physical difference between this particle and a free particle. A particle with total energy $E_1 = K + c$ in a constant potential energy $U_1(x) = c$ is the physically identical to a particle with total energy $E_2 = K$ in a vanishing potential energy $U_2(x) = 0$. For both cases the Schrödinger equation gives the correct wavelength.

2. Let $\Psi(x)$ be a solution of the one-dimensional Schrödinger equation for a given potential $U(x)$ with energy E. Show that the function $\widehat{\Psi}(x) = A\Psi(x)$ is also a solution of the one dimensional Schrödinger equation with potential $U(x)$, with the same E, for any values of the constant A. What does this show about normalizing a wave function?

The Schrödinger equation for a particle with mass m in a potential $U(x)$ is:

$$\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2}(E-U)\Psi = 0\ ,$$

where E is the energy of the solution $\Psi(x)$. The wave function $\hat{\Psi}(x) = A\Psi(x)$ is also a solution of this equation with the same energy E, if

$$\frac{d^2\hat{\Psi}}{dx^2} + \frac{2m}{\hbar^2}(E-U)\hat{\Psi} = 0\ .$$

Some calculus gives:

$$A\left(\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2}(E-U)\Psi\right) = 0\ .$$

and that holds since Ψ was a solution of the Schrödinger equation with energy E.

Thus: normalization of a wave function does not change the corresponding energy!

3.

a. Explicitly show by differentiation that $\psi(x) = Ae^{kx} + Be^{-kx}$ solve the Schrödinger equation for a region where $V = V_0 < \infty$ (a constant) and $E < V_0$. How is the variable k defined?
b. Suppose we would take the $\lim_{V_0 \to \infty} V_0$, what does this imply for k?

a) The Schrödinger equation reads in this case

$$\frac{-\hbar^2}{2m}\frac{d^2}{dx^2}\psi + V_0\psi = E\psi \Rightarrow -\frac{d^2\psi}{dx^2} = \frac{2m\overbrace{(E-V)}^{<0}}{\hbar^2}\psi \Rightarrow \frac{d^2\psi}{dx^2} = \frac{2m(V-E)}{\hbar^2}\psi \Rightarrow \frac{d^2\psi}{dx^2} = k^2\psi,$$

$$k = \sqrt{\frac{2m(V-E)}{\hbar^2}}$$

Now we know our k. Differentiating the proposed wave function twice yields:

$$\frac{d^2\psi}{dx^2} = k^2 A\exp(kx) + (-k)(-k)B\exp(-kx) = k^2\left(A\exp(kx) + B\exp(-kx)\right) = k^2\psi$$

And thus it clearly solves the Schrödinger equation.

b) If V_0 is infinity then that means that k is infinite as well for any x. The function $\exp(kx)$ is then blown up for any x so it is not normalizable. The function $\exp(-kx) = 0$ for all x since we take the limit of $\lim_{y\to\infty}\exp(-y) = 0$ for all x. In informal words, no matter how small x: k wins and takes the argument of the exponent to infinity in magnitude.

4.

a. Explicitly show by differentiation that both $\psi_a(x) = Acos(kx) + Bsin(kx)$ and $\psi_b(x) = Ce^{ikx} + De^{-ikx}$ solve the Schrodinger equation for a region where $V = V0$ (a constant) and $E > V0$. How is the variable k defined?
b. Use Eulers identity for complex exponentials to show that $\psi_a(x)$ and $\psi_b(x)$ are equivalent solutions by relating A, B, C and D. Please remember that A, B, C and D are in general complex coefficients.

a) The Schrödinger equation reads in this case

$$\frac{-\hbar^2}{2m}\frac{d^2}{dx^2}\psi + V_0\psi = E\psi \Rightarrow -\frac{d^2\psi}{dx^2} = \frac{2m\overbrace{(E-V)}^{>0}}{\hbar^2}\psi \Rightarrow \frac{d^2\psi}{dx^2} = -k^2\psi,$$
$$k = \sqrt{\frac{2m(E-V)}{\hbar^2}}$$

Now we know our k. Differentiating the proposed wave functions twice yields:

$$\frac{d^2\psi}{dx^2} = i^2k^2C\exp(ikx) + (-ik)(-ik)D\exp(-ikx) = i^2k^2\left(C\exp(ikx) + D\exp(-ikx)\right) = -k^2\psi$$

and

$$\frac{d^2\psi}{dx^2} = \frac{d}{dx}\left(-kA\sin(kx) + kB\cos(kx)\right) = -k^2A\cos(kx) - k^2B\sin(kx) = -k^2\psi$$

And thus they both clearly solve the Schrödinger equation.

b) We use the Euler identity and work out the product.

$$\begin{aligned} C\exp(ikx) + D\exp(-ikx) &= C(\cos(kx) + i\sin(kx)) + D(\cos(kx) - i\sin(kx)) \\ &= (C+D)\cos(kx) + i(C-D)\sin(kx) \end{aligned}$$

So this means that if we write the solution as $C\exp(ikx) + D\exp(-ikx)$, it is the same as writing $A\cos(kx) + B\sin(kx)$ with $A = C + D$ and $B = i(C - D)$.
Equivalently, one could demonstrate:

$$\begin{aligned} A\cos(kx) + B\sin(kx) =& A\frac{\exp(ikx) + \exp(-ikx)}{2} + B\frac{\exp(ikx) - \exp(-ikx)}{2i} = \\ & A\frac{\exp(ikx) + \exp(-ikx)}{2} - iB\frac{\exp(ikx) - \exp(-ikx)}{2} = \\ & \frac{A - iB}{2}\exp(ikx) + \frac{A + iB}{2}\exp(-ikx) \end{aligned}$$

5. For the one dimensional infinite square well with boundaries set at $x = 0$ and $x = a$ the wave function corresponding to the n'th stationary state is given by the following function:

$$\psi(x) = \begin{cases} \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi x}{a}\right), & x \in [0, a] \\ 0 \quad , & x < 0 \quad \text{or} \quad x > a \end{cases}$$

   a. Compute the probability density for all the allowed states.
   b. For $n = 1, 2, 3$ plot the probability density for the particle. Afterwards give interpretations of the plots! (If you find this difficult without numerical values, you may used $a = 5.29 \cdot 10^{-11}\text{m}$ (= Bohr radius to 3 significant digits)
   c. Compute the expected value of position for a particle in eigenstate n. Give your physical interpretation of this result.
      Remember: $\langle x \rangle = \int \psi^*(x)\, x\, \psi(x)\, dx$

Hint for calculation: Substitute $y = \frac{n\pi x}{a}$ *and use* $\sin^2(y) = \frac{1-\cos(2y)}{2}$. *Afterwards use partial integration to reduce the integral to an integral of an elementary function.*

(a) The probability density is the square of the in general complex wavefunction: $|\psi|^2 = \psi^*\psi$. Since the wave function is real in this case $\psi^* = \psi$ and thus:

$$|\psi|^2 = \left(\sqrt{\frac{2}{a}}\sin\left(\frac{n\pi x}{a}\right)\right)^2 = \frac{2}{a}\sin^2\left(\frac{n\pi x}{a}\right)$$

(b) The interpretation is as follows: The probability to find a particle at a certain x, depends on the energy level E through the relation $E = \frac{n^2\pi^2\hbar^2}{2ma^2}$. In general it is more probable to measure the position of the particle being closer to the box's edges at higher energy levels, but for $n = 1, 3$ (and in general all odd) there is a 'lobe' in the middle of the box, which decreases in width for higher n. That lobe is absent for the even quantum numbers n. The vertical red lines are the calculated values for exercise 5c.

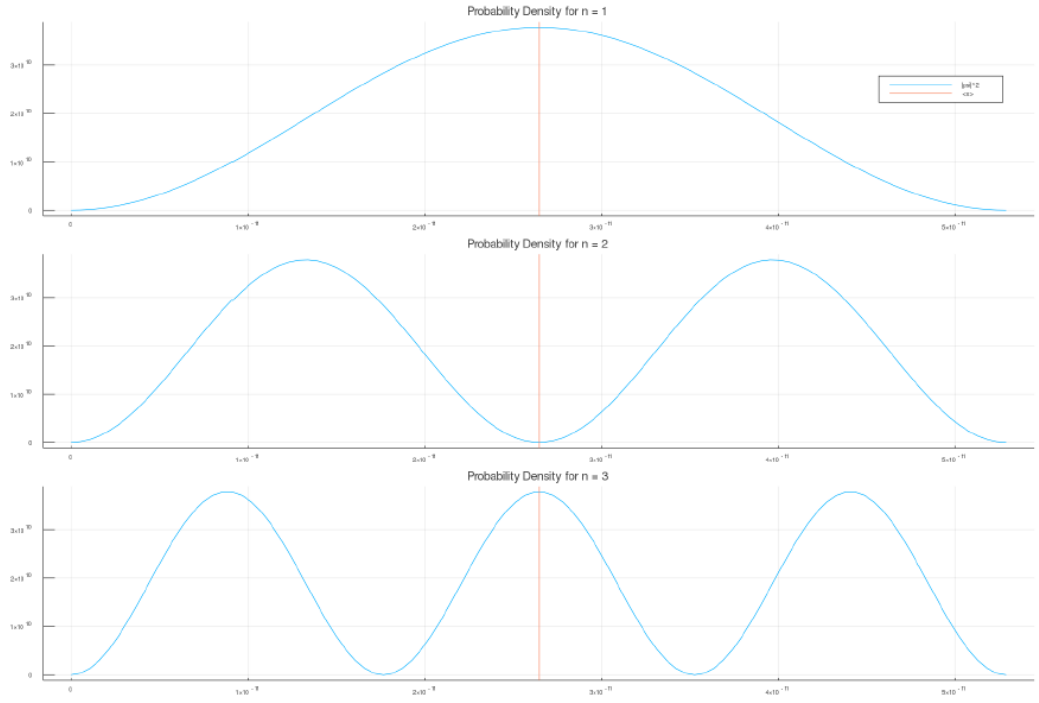


Figure 1: Plots of the probability Densities.

(c) *Calculation:*
Start with plugging in the definition and combining terms:

$$< x >= \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \cdot x \cdot \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \, dx = \frac{2}{a} \int_0^a x \sin^2\left(\frac{n\pi x}{a}\right)$$

Define $y = \frac{n\pi x}{a} \longrightarrow \frac{dy}{dx} = \frac{n\pi}{a} \longrightarrow dx = \frac{a}{n\pi} dy$, while also using that $y(x = 0) = 0$ and $y(x = a) = n\pi$ yields:

$$\int_0^a x \sin^2\left(\frac{n\pi x}{a}\right) = \frac{2}{a} \int_{y(0)}^{y(a)} \frac{ay}{n\pi} \sin^2(y) \frac{a}{n\pi} dy = \frac{2a}{n^2\pi^2} \int_0^{n\pi} y \sin^2(y) dy$$

Use the hint to rewrite the integral to reduce the order of the trigonometric function and solve using partial integration

$$\frac{2a}{n^2\pi^2} \int_0^{n\pi} y \sin^2(y) dy = \frac{2a}{n^2\pi^2} \int_0^{n\pi} y \left(\frac{1 - \cos(2y)}{2}\right) dy = \frac{2a}{n^2\pi^2} \left(\frac{y^2}{4}\Big|_0^{n\pi} - \frac{1}{2} \int_0^{n\pi} y \cos(2y)\right)$$

$$= \frac{2a}{n^2\pi^2} \left(\frac{y^2}{4}\Big|_0^{n\pi} - \frac{1}{4} y \sin(2y)\Big|_0^{n\pi} + \frac{1}{4} \int_0^{n\pi} \sin(2y)\right) = \frac{2a}{n^2\pi^2} \left(\frac{y^2}{4}\Big|_0^{n\pi} - \frac{1}{4} y \sin(2y)\Big|_0^{n\pi} - \frac{1}{8} \cos(2y)\Big|_0^{n\pi}\right)$$

$$= \frac{2a}{n^2\pi^2} \left(\frac{n^2\pi^2}{4} - 0 + 0\right) = \frac{a}{2}$$

Note that the the expected value of the position does NOT depend on the state n. *Interpretation:*
Taking the 10000 identical experiments as an example, one expects that the results of these measurement will yield different outcomes. If you average these measurements, the result is the calculated $< x >$ independent of the energy that we prepared the particles in. In this case that is in the middle of the box, which makes sense due to the potential being symmetric with respect to that point.

6. Compute in detail the wave functions and corresponding energies for a two-dimensional rectangular box with size L_x by L_y. Give the expression and calculate the normalization constant. Use this calculation to determine the wave function for a square box with side length L.

The two-dimensional Schrödinger equation in a potential $U(x,y)$ is given by

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\Psi(x,y)}{\partial x^2}+\frac{\partial^2\Psi(x,y)}{\partial y^2}\right)+U(x,y)\Psi(x,y)=E\Psi(x,y)\ ,$$

with E the total energy of the solution. Note that $\Psi(x,y)$ is now a function of two variables, and the Schrödinger equation is now a partial differential equation. Suppose the box is from $x=0$ to $x=L_x$ and from $y=0$ to $y=L_y$. Since the potential $U(x,y)=\infty$ outside the box, we have $\Psi(x,y)=0$ outside the box. Inside the box we obtain the equation

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\Psi(x,y)}{\partial x^2}+\frac{\partial^2\Psi(x,y)}{\partial y^2}\right)=E\Psi(x,y)\ ,$$

To find a solution we try separation of variables: $\Psi(x,y)=f(x)g(y)$. Substitution in the Schrödinger equation inside the box gives

$$-\frac{\hbar^2}{2m}\left(\frac{d^2f(x)}{dx^2}g(y)+f(x)\frac{d^2g(y)}{dy^2}\right)=Ef(x)g(y)\ .$$

Division by $f(x)g(y)$ gives

$$\frac{1}{f(x)}\frac{d^2f(x)}{dx^2}+\frac{1}{g(y)}\frac{d^2g(y)}{dy^2}=-\frac{2mE}{\hbar^2}\ .$$

Note that $\frac{1}{f(x)}\frac{d^2f(x)}{dx^2}$ is a function of only x and $\frac{1}{g(y)}\frac{d^2g(y)}{dy^2}$ is a function of only y. Their sum can only be constant (for all x and y in the box) if both parts are constant. Hence we have

$$\frac{1}{f(x)}\frac{d^2f(x)}{dx^2}=-k_x^2,\qquad \frac{1}{g(y)}\frac{d^2g(y)}{dy^2}=-k_y^2\ .$$

(We take both constants in the right hand side negative, since positive constants will give exponential solutions that will not satisfy the boundary conditions.) The values of k_x and k_y are related to the energy E by

$$k_x^2+k_y^2=2mE/\hbar^2 \tag{1}$$

The general solutions of the first equation are

$$f(x)=A\sin(k_xx)+B\cos(k_xx)\ .$$

The boundary condition (continuity of $\Psi(x,y)=f(x)g(y)$) in $x=0$ implies that $B=0$. The boundary condition in $x=L_x$ implies that $k_xL_x=n_x\pi$ for $n_x=1,2,3,\ldots$.

The general solutions of the second equation are

$$g(y)=C\sin(k_yy)+D\cos(k_yy)\ .$$

The boundary condition (continuity of $\Psi(x,y)=f(x)g(y)$) in $y=0$ implies that $D=0$. The boundary condition in $y=L_y$ implies that $k_yL_y=n_y\pi$ for $n_y=1,2,3,\ldots$.

Hence we can select two quantum numbers n_x and n_y, with solution inside the box

$$\Psi_{n_x,n_y}(x,y)=AC\sin\left(\frac{n_x\pi x}{L_x}\right)\sin\left(\frac{n_y\pi y}{L_y}\right)\ .$$

Outside the box always $\Psi_{n_x,n_y}(x,y)=0$. From (1) we obtain that the corresponding energy values are given by

$$E_{n_x,n_y}=\frac{\hbar^2}{2m}\left(\frac{n_x^2\pi^2}{L_x^2}+\frac{n_y^2\pi^2}{L_y^2}\right)\ .$$

To normalize the wave function, we require that

$$\iint_{\text{box}} \Psi^2_{n_x,n_y}(x,y)dxdy = 1.$$

This yields

$$A^2C^2 \int_0^{L_x} \sin^2\left(\frac{n_x\pi x}{L_x}\right)dx \int_0^{L_y} \sin^2\left(\frac{n_y\pi y}{L_y}\right)dy = A^2C^2\frac{L_x}{2}\frac{L_y}{2} = 1\ .$$

This implies that $AC = \dfrac{2}{\sqrt{L_xL_y}}$. Hence the normalized wave functions are

$$\Psi_{n_x,n_y}(x,y) = \frac{2}{\sqrt{L_xL_y}} \sin\left(\frac{n_x\pi x}{L_x}\right)\sin\left(\frac{n_y\pi y}{L_y}\right)\ .$$

Note that there are many degenerate energy levels, for instance $n_x = 1, n_y = 7$ gives the same energy as $n_x = 5, n_y = 5$ or as $n_x = 7, n_y = 1$.

The case of a rectangular square box with sides L can simply be computed from the general expression by setting $L_x = L_y = L$.

$$\Psi_{n_x,n_y}(x,y) = \frac{2}{L}\ \sin\left(\frac{n_x\pi x}{L}\right)\sin\left(\frac{n_y\pi y}{L}\right)\ ,$$

and corresponding energy levels

$$E_{n_x,n_y} = \frac{\hbar^2}{2m}\left(\frac{n_x^2\pi^2}{L^2} + \frac{n_y^2\pi^2}{L^2}\right) = \frac{\hbar^2}{2m}\left(\frac{(n_x^2+n_y^2)\pi^2}{L^2}\right)\ .$$

7. A particle is represented by the following wave function:

$$\psi(x) = \begin{cases} 0, & x < -\dfrac{L}{2} \\ C\left(\dfrac{2x}{L}+1\right), & -\dfrac{L}{2} < x < 0 \\ C\left(\dfrac{2x}{L}+1\right), & 0 < x < \dfrac{L}{2} \\ 0, & x > \dfrac{L}{2} \end{cases}$$

   a. Use the normalization condition to find C.
   b. Evaluate the probability to find the particle in an interval of width $0.010L$ at $x = L/4$ (that is, between $x = 0.245L$ and $x = 0.255L$. (No integral is necessary for this calculation.)
   c. Evaluate the probability to find the particle between $x = 0$ and $x = +\frac{L}{4}$.
   d. Find the average value of x and the rms-value of x: $x_{rms} = \sqrt{(x^2)_{av}}$

(a) The regions with $x < -L/2$ and $x > +L/2$ do not contribute to the normalization integral. The remaining integral is:

$$\int |\psi(x)|^2\, dx = \int_{-L/2}^{0} C^2(2x/L+1)^2\, dx + \int_{0}^{L/2} C^2(-2x/L+1)^2\, dx$$

$$= C^2\,(4x^3/3L^2 + 2x^2/L + x)\Big|_{-L/2}^{0} + C^2\,(4x^3/3L^2 - 2x^2/L + x)\Big|_{0}^{L/2} = C^2 L/3$$

Setting the integral equal to 1 gives $C = \sqrt{3/L}$.

(b) $P(x)\,dx = |\psi(x)|^2\, dx = C^2(4x^2/L^2 - 4x/L + 1)\, dx$ and with $x = 0.250L$ and $dx = 0.010L$,

$$P(x)\,dx = \frac{3}{L}\left(\frac{4}{L^2}\frac{L^2}{16} - \frac{4}{L}\frac{L}{4} + 1\right)(0.010L) = 0.0075$$

(c)

$$P(0 : L/4) = \int_0^{L/4} |\psi(x)|^2\, dx = C^2 \int_0^{L/4}\left(\frac{4}{L^2}x^2 - \frac{4}{L}x + 1\right)dx = \frac{3}{L}\left(\frac{4}{L^2}\frac{x^3}{3} - \frac{4}{L}\frac{x^2}{2} + x\right)\Bigg|_0^{L/2} = \frac{1}{2}$$

(d) $$\langle x\rangle = \int |\psi(x)|^2\, x\, dx = \frac{3}{L}\int_{-L/2}^{0}\left(\frac{4x^2}{L^2} + \frac{4x}{L} + 1\right) x\, dx + \int_0^{L/2}\left(\frac{4x^2}{L^2} - \frac{4x}{L} + 1\right) x\, dx$$

$$= \frac{3}{L}\left(\frac{x^4}{L^2} + \frac{4x^3}{3L} + \frac{x^2}{2}\right)\Bigg|_{-L/2}^{0} + \frac{3}{L}\left(\frac{x^4}{L^2} - \frac{4x^3}{3L} + \frac{x^2}{2}\right)\Bigg|_{0}^{L/2} = 0$$

It is apparent from the shape of the wave function that the equal probability densities for positive and negative x cancel to give an average of zero.

$$\langle x^2\rangle = \int |\psi(x)|^2\, x^2\, dx = \frac{3}{L}\int_{-L/2}^{0}\left(\frac{4x^2}{L^2} + \frac{4x}{L} + 1\right) x^2\, dx + \int_0^{L/2}\left(\frac{4x^2}{L^2} - \frac{4x}{L} + 1\right) x^2\, dx$$

$$= \frac{3}{L}\left(\frac{4x^5}{5L^2} + \frac{x^4}{L} + \frac{x^3}{3}\right)\Bigg|_{-L/2}^{0} + \frac{3}{L}\left(\frac{4x^5}{5L^2} - \frac{x^4}{L} + \frac{x^3}{3}\right)\Bigg|_{0}^{L/2} = \frac{L^2}{40}$$

The rms value is then $x_{\text{rms}} = \sqrt{\langle x^2\rangle} = \sqrt{L^2/40} = 0.158L$.

8. In this exercise we consider a particle with mass m in a potential $U(x)$ given by

$$U(x) = \begin{cases} \infty & \text{if} \quad x < 0 \\ 0 & \text{if} \quad 0 \le x \le L \\ U_0 & \text{if} \quad x > L \end{cases}$$

where U_0 is some positive number, see also Figure 1. Hence at the left wall of the well (at $x = 0$) the potential goes from 0 to ∞ and at the right wall (at $x = L$) the potential goes from 0 to U_0.

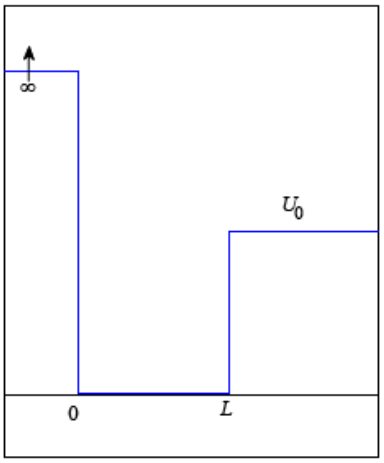


Figure 1: The potential $U(x)$.

In this exercise we try to find solutions to the Schrödinger equation for this situation with a total energy E with $0 < E < U_0$.

a. Show that in classical ("Newton") mechanics a particle with total (=potential + kinetic) energy $E < U_0$ cannot escape from the well.

Answer: In classical mechanics the total energy of a particle with mass m, velocity v and position x is $E = \frac{1}{2}mv^2 + U(x)$. Since the kinetic energy $\frac{1}{2}mv^2$ is always ≥ 0, the total energy E of a particle can never be smaller than the potential energy $U(x)$ at the current position x. To escape from the well (to the right) the particle must have a potential energy U_0, which is impossible with a total energy $E < U_0$.

b. What is the solution $\Psi(x)$ of the Schrödinger equation for $x < 0$?

Answer: For $x < 0$ the potential is $+\infty$, hence $\Psi(x) = 0$ for $x < 0$.

c. What is the general solution $\Psi(x)$ of the Schrödinger equation for $0 \le x \le L$?

Answer: In this region $U(x) = 0$, hence the Schrödinger equation is

$$\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2}E\Psi = 0 \ ,$$

with general solution

$$\Psi(x) = A\sin(kx) + B\cos(kx) \quad \text{where} \quad k^2 = \frac{2mE}{\hbar^2} \ .$$

d. Show that continuity of $\Psi(x)$ in x = 0 implies that the solution for $0 \le x \le L$ must be of the form $\Psi(x) = A\sin kx$.

Answer: Since for $x < 0$ $\Psi(x) = 0$ we must have $\Psi(0) = A\sin(0) + B\cos(0) = 0$, which implies $B = 0$.

e. Give k^2 in terms of the energy E.

Answer:

$$k^2 = \frac{2mE}{\hbar^2} .$$

f. What is the general solution $\Psi(x)$ of the Schrödinger equation for $x > L$?

Answer: In this region $U(x) = U_0$, hence the Schrödinger equation is

$$\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2}(E - U_0)\Psi = 0 ,$$

Since we look for solutions with $0 < E < U_0$, we rewrite the equation as

$$\frac{d^2\Psi}{dx^2} = a^2\Psi \quad \text{with} \quad a = \sqrt{\frac{2m}{\hbar^2}(U_0 - E)} > 0 .$$

The general solution for $x > L$ is

$$\Psi(x) = Ce^{-ax} + De^{ax} .$$

g. Show that the behavior for $x \to \infty$ requires that the solution for $x > L$ is of the form $\Psi(x) = Ce^{-ax}$ with $a > 0$.

Answer: Since $a > 0$ the term with e^{ax} explodes for $x \to \infty$. Hence D must be 0.

h. Give a^2 in terms of the energy E and the potential value U_0.

Answer:

$$a^2 = \frac{2m}{\hbar^2}(U_0 - E) .$$

i. The continuity of $\Psi(x)$ and of $\frac{d\Psi(x)}{dx}$ in $x = L$ leads to two conditions. Give these conditions.

Answer: In general $\Psi(x)$ and the derivative $d\Psi(x)/dx$ must be continuous. (Only at places where the potential "jumps" to ∞ (like $x = 0$ in this exercise) only $\Psi(x)$ must be continuous.) The continuity of $\Psi(x)$ in $x = L$ gives the condition

$$A\sin(kL) = Ce^{-aL} .$$

The continuity of the derivative $d\Psi(x)/dx$ in $x = L$ leads to the condition

$$Ak\cos(kL) = -Cae^{-aL} .$$

j. Show that part (i) implies that k and a must be related by

$$a = -k\frac{\cos kL}{\sin kL} \qquad (1)$$

Answer: By dividing the two conditions given above we obtain

$$\frac{Ak\cos(kL)}{A\sin(kL)} = \frac{-Cae^{-aL}}{Ce^{-aL}} ,$$

k. Show that parts (e) and (h) imply that a and k also must satisfy the condition

$$a = \sqrt{\widehat{U_0} - k\text{^}2} \qquad (2)$$

And give $\widehat{U_0}$.

Answer: Using $k^2 = \frac{2mE}{\hbar^2}$ and $a = \sqrt{\frac{2m}{\hbar^2}(U_0 - E)}$ we obtain

$$a = \sqrt{\frac{2m}{\hbar^2}(U_0 - E)} = \sqrt{\frac{2m}{\hbar^2}U_0 - \frac{2m}{\hbar^2}E} = \sqrt{\widehat{U_0} - k^2}\ ,$$

with $\widehat{U_0} = \frac{2mU_0}{\hbar^2}$.

A solution $\Psi(x)$ of the given form exists if there are a and k values that satisfy the two equations (1) and (2). To find out whether solutions exist, we first introduce new (dimensionless) variables x and y by $x = kL$ and $y = aL$.

l. Show that in terms of the new unknowns and y the equations are

$$y = -x\frac{\cos x}{\sin x}$$

$$y = \sqrt{F^2 - x^2}$$

With $F^2 = \frac{2mU_0L^2}{\hbar^2}$.

Answer: Substitution of $x = kL$ and $y = aL$ in (2) gives

$$\frac{y}{L} = -\frac{x}{L}\frac{\cos(x)}{\sin(x)}\ .$$

Multiplication by L gives the required relation. To prove the second relation we multiply (3) by L:

$$La = \sqrt{L^2\widehat{U_0} - L^2k^2}\ .$$

m. The graph of the function $y = -x\frac{\cos x}{\sin x}$ is the red line in Figure 2. To find the intersections with the graph $y = \sqrt{F^2 - x^2}$ recall that the last graph is the supper part of a circle with radius F. How many solutions are there for $F = 3$? And for $F = 6$? *Hint: use a compass*

Answer: The graphs of $y = \sqrt{F^2 - x^2}$ are the green circles shown in Figure 4 for $F = 3$, $F = 6$ and $F = 9.9821$. The smallest circle, with radius 3, has only one intersection with a red line. Hence for $F = 3$ there is one solution. That means that for $U_0 = \hbar^2F^2/(2mL^2) = 9\hbar^2/(2mL^2)$ there is only one possible wave function Ψ. For $F = 6$, i.e. $U_0 = \hbar^2F^2/(2mL^2) = 18\hbar^2/(mL^2)$ there are two possible wave functions $\Psi(x)$.

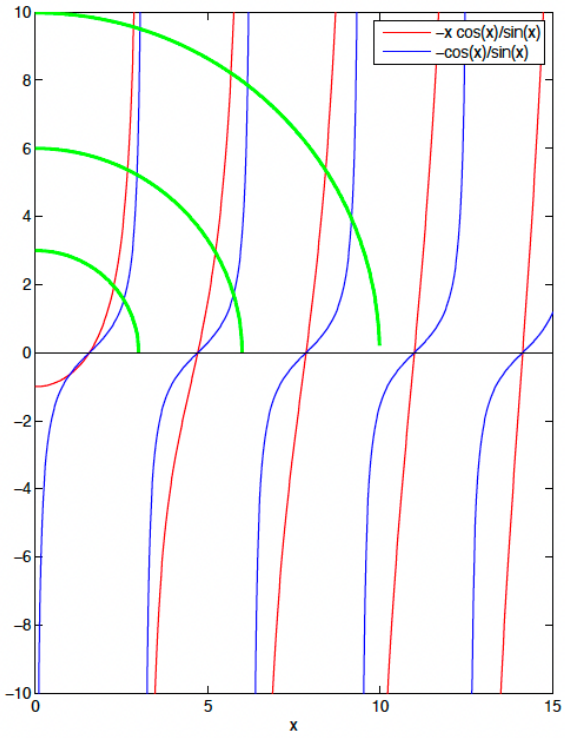


Figure 4: Exercise 6: the graphs of $y = -\cos(x)/\sin(x)$ (blue) and $y = -x\cos(x)/\sin(x)$ (red) and the circles with center $(0,0)$ and radius 3, 6, 9.9812 (green)

n. Suppose the particle is an electron, $L = 1.0 \cdot 10^{-10}$m and $U_0 = 380$ eV. How many discrete energy levels are there is this case?

Answer: Computation of F for this case yields $F = \sqrt{\frac{2mU_0L^2}{\hbar^2}} = 9.9821$. This corresponds to the largest green circle, which has three intersections with the red lines. Hence in this situation there are three possible energy values.

As illustration we give in Figure 5 the wave functions $\Psi(x)$ for the three considered cases, using $L = 1.0 \cdot 10^{-10}$ m. Note that, for $0 \leq x \leq L$ (inside the well) the wave function $\Psi(x)$ has an oscillatory behavior, while for $x > L$ (to the right of the well) $\Psi(x)$ is a decaying exponential. Computing the wave functions was not part of the exercise.

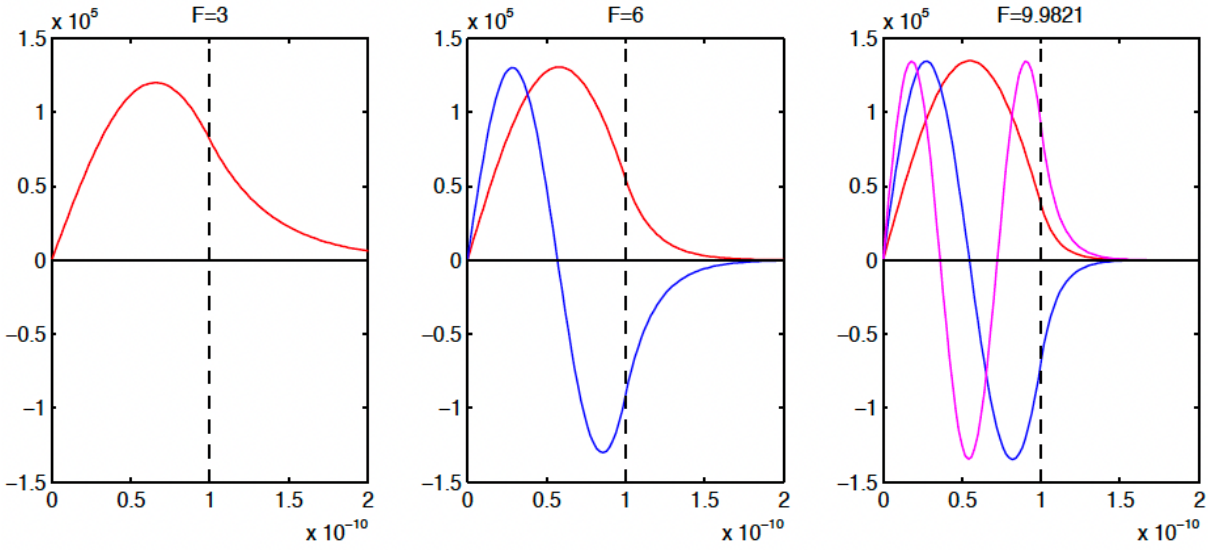


Figure 5: Exercise 6: the wave functions $\Psi(x)$ for the considered cases, $L = 10^{-10}$ m